<u>ASSIGNMENT-2</u>

IMPORTANT INSTRUCTIONS

- *All questions in this assignment are coding questions.*

SUBMISSION FORMAT:

- *Create one Notebook (.ipynb) file*
- *Solve all four questions in this single notebook.*
- *Add clear markdown cells with question titles.*
- *Include comments in your code to explain it wherever needed.*
- *Upload the completed (.ipynb) file to Google Classroom*

For queries, contact us on Whatsapp.

Question 1: Expected Value and Variance in Particle Motion

A physicist measures the speeds of nitrogen gas particles at room temperature. The recorded speeds (in m/s) are: 420, 445, 380, 510, 398, 475, 425, 390, 460, 440. Write codes for the following tasks:

Part (a): Calculate the average speed and store it in a variable called avg_speed. Print the result with the message "Average speed = X m/s".

Part (b): Calculate the standard deviation of the speeds and store it in a variable called std_speed. Print the result with the message "Standard deviation = X m/s".

Part (c): Create a bar plot where the x-axis shows the particle number (1 through 10) and the y-axis shows the speed of each particle. Add a horizontal line at the average speed position.
Label the axes as "Particle Number" and "Speed (m/s)". Add a title "Particle Speed Distribution with Mean".

Question 2: Normal Distribution and Probability Calculation

A temperature sensor takes 100 measurements of room temperature. The measurements follow a normal distribution with mean 22°C and standard deviation 0.5°C. Write codes for the following tasks:

Part (a): Generate 100 random temperature measurements with mean 22 and standard deviation 0.5. Store these values in an array called temperatures.

Part (b): Create a histogram with 20 bins showing the distribution of temperatures. Label the x-axis as "Temperature (°C)" and the y-axis as "Frequency". Add a title "Distribution of Temperature Measurements".

Part (c): Write code to count how many measurements fall between 21.5°C and 22.5°C. Write code to calculate the percentage of measurements in this range by dividing the count by 100 and multiplying by 100. Print the result with the message "Percentage of measurements between 21.5°C and 22.5°C = X%".

Question 3: Law of Large Numbers

A physicist generates random velocities of gas particles. You need to show that averaging many particles gives a stable result. Write codes for the following tasks:

Part (a): Generate 5000 random velocity values with scale parameter 2.5. Store these in an array called velocities.

Part (b): Calculate the average of the first 100 velocities, the first 1000 velocities, the first 3000 velocities, and all 5000 velocities. Create a list called averages storing these four values in order.

Part (c): Create a line plot with a list of sample sizes on the x-axis and the averages on the y-axis. Label the x-axis as "Number of Samples" and the y-axis as "Average Velocity". Add a title "Law of Large Numbers: Convergence of Average". Add a horizontal dashed red line at the final average value (the 5000-sample average).

[Hint: using plt.axhline() with linestyle='--' and color='red'.]

Question 4: Comparing Two Distributions

A metal sample is measured by two different instruments. Instrument 1 gives precise measurements centered at 5.80 with standard deviation 0.15. Instrument 2 gives scattered measurements centered at 5.75 with standard deviation 0.35. Write codes for the following tasks:

Part (a): Generate 500 measurements from Instrument 1 with mean 5.80 and std 0.15. Store in array inst1. Generate 500 measurements from Instrument 2 with mean 5.75 and std 0.35. Store in array inst2.

Part (b): On a single plot, create two overlapping histograms with 15 bins each. Plot inst1 with color 'blue' and inst2 with color 'red'. Label the x-axis as "Conductivity (S/m)" and the y-axis as "Frequency". Add a legend with labels 'Instrument 1' and 'Instrument 2'. Add a title "Comparison of Two Instruments".

(Hint: Use the alpha parameter inside plt.hist() to make the bars transparent so you can see both histograms where they overlap. For example, use alpha=0.6. An alpha of 1 is solid, and 0 is invisible.)

Part (c): Calculate the ratio of the standard deviation of inst2 to the standard deviation of inst1. Store this ratio in a variable called precision_ratio. Print the result with the message "Instrument 2 is X times more scattered than Instrument 1" (where X is the precision_ratio rounded to two decimal places).